

DEPRESSION INDICATOR PREDICTION

Using NHANES 2017-MAR 2020 & 2021-2023 Data



Group 3: Angela Johnson, Annabel Tapia, Naeem Chaudhry, Thiri Kyi



BACKGROUND

- Depression is a common mental health condition, it affects mood, energy, sleep, and daily functioning for weeks or longer.
- It can seriously impact physical health, relationships, work, and overall quality of life.
- Doctors and researchers screen for it using standardized questionnaires, our project uses one of the most common ones, the PHQ-9.



OUR GOAL

Explore whether everyday health and lifestyle information can help flag higher depression risk.

WHY IT MATTERS

13.1%

of U.S. adolescents & adults
screened positive for depression
(CDC, NHANES 2021–2023)

60%

increase in depression
prevalence over the past decade
(CDC, released 2025)

- Depression is common, disabling, and treatable, but many cases go unnoticed.
- Spotting patterns in everyday health and lifestyle data could support earlier screening conversations.
- This is a chance for our team to apply real machine learning skills to a meaningful public health question.

THE DATASET

DATASET AND PREDICTION

Source: NHANES, combining two cycles 2017-March 2020 (pre-pandemic) and 2021–2023 (post-pandemic) for a larger robust sample

Target: a Depression Flag, built from the PHQ-9 (9 standard screening questions)

Score of 10 or higher → flagged as likely depression.
Below that, not flagged.

13,731

people with a usable
PHQ-9 score

10.9%

flagged for likely
depression (1,490 people)

The 2021-2023 slice alone (13.3%) closely matches CDC's national estimate for that period (13.1%) and the rise from the pre-pandemic cycle (9.3%) mirrors CDC's reported decade-long increase.

WHAT WE ARE USING TO PREDICT



DEMOGRAPHICS

- Age, gender
- Education
- Marital status
- Born in the US



LIFESTYLE

- Sleep hours (weekday/weekend)
- Physical activity minutes
- Smoking and alcohol use
- HIV testing history



INCOME

- Income-to-poverty ratio
- Monthly poverty index
- Family income category

GETTING THE DATA READY

- Raw survey data uses special codes for “Refused” and “Don’t know”, we have converted those to real missing values
- Combined two separate NHANES survey cycles into one consistent table on a shared set of columns
- Filled in the small number of remaining gaps so that every row was complete

BEFORE:

27,493

people across two survey
cycles

AFTER:

13,731

people across two survey
cycles

BEFORE:

116-118

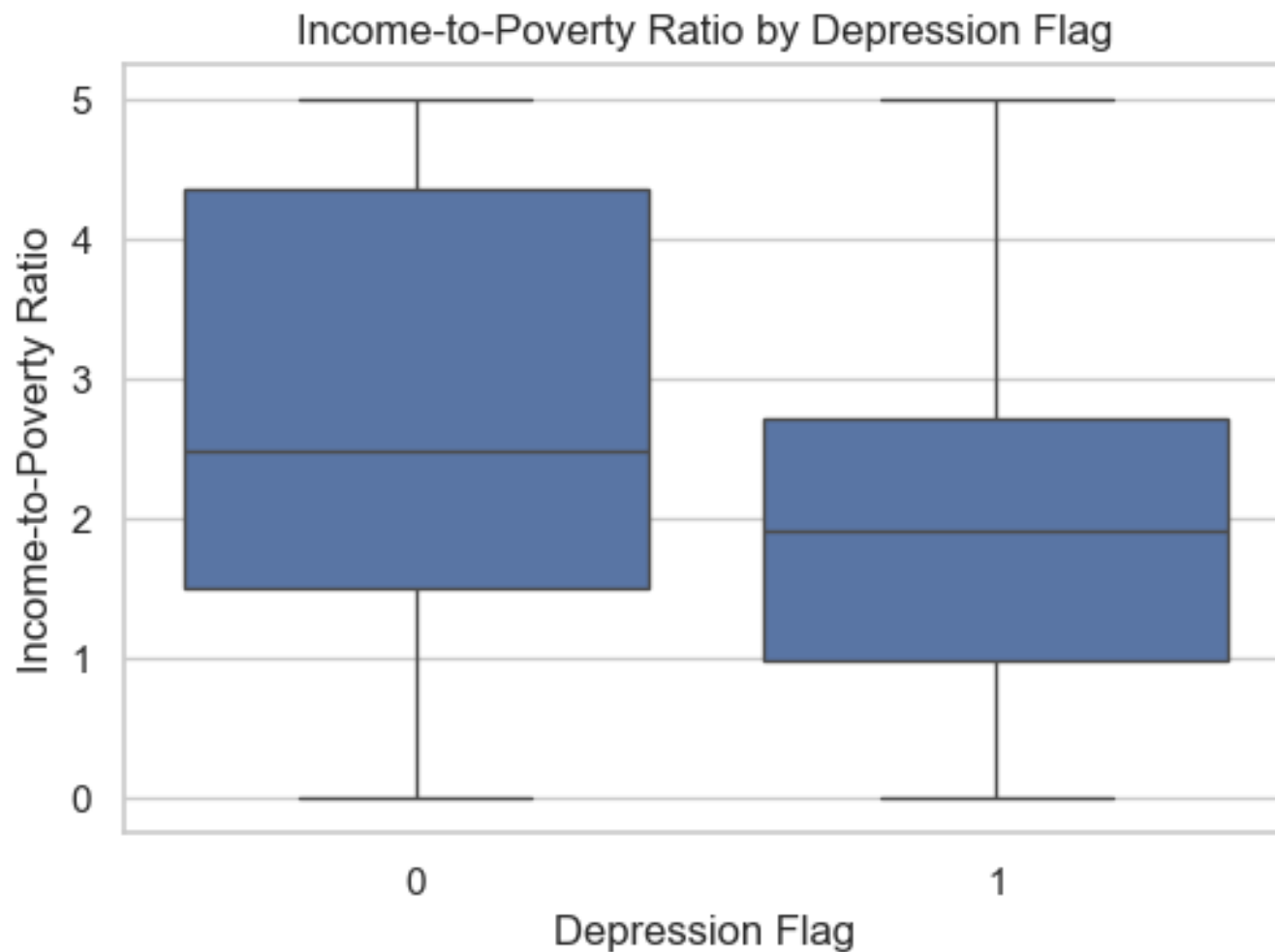
raw columns

AFTER:

28

clean columns, 0 missing

EARLY PATTERNS



WHAT STANDS OUT:

- Participants flagged for depression tended to have lower income-to-poverty ratios.
- The average was 2.13 for the flagged group, compared with 2.79 for the group not flagged.
- The overlap shows that income is only one part of depression risk.

IMPLEMENTATION AND TOOLS

Core stack

- Python, Jupyter Notebook
- Pandas, NumPy
- GitHub for version control
- Matplotlib

Modeling Libraries

- Scikit-learn (Logistic Regression, SVM, RF)
- CatBoost, XGBoost

All libraries are third-party (open source)

WHAT WE ARE USING TO PREDICT

- We are framing this binary classification: predict Depression_Flag (0 or 1) from the cleaned features
- Everyone trains and tests on the exact same data split giving us results that are comparable
- Only ~11% of cases are flagged; we are addressing this class imbalance rather than relying on plain accuracy

CatBoost

Angela Johnson

Random Forest & Logistic Regression

Thiri Kyi

SVM

Naeem Chaudhry

XGBoost

Annabel Tapia

RESULTS

Model	Accuracy	Precision	Recall	F1-score
CatBoost	0.7573	0.2257	0.4638	0.3036
Random Forest	0.7920	0.2232	0.3277	0.2655
Logistic Regression	0.7820	0.2415	0.4255	0.3082
SVM	0.7864	0.2577	0.4638	0.3313
XGBoost	0.7723	0.2365	0.4468	0.3093

LIVE DEMO

CONCLUSION

ACCOMPLISHMENTS:

- Built a full reproducible cleaning pipeline that combines two NHANES cycles into one model-read dataset
- Our data's rate (13.3% post-pandemic vs 9.3 pre-pandemic) echoes CDC's own reported national trend
- Checked whether our top model changed depending on the metric used and reported it honestly instead of picking whichever looked best
- Verified all five models were compared fairly, with no data leakage in any of them

CHALLENGES:

- Five of our originally planned predictors BMI, employment, general health, chronic disease history, and healthcare access were never available in the NHANES files we could access, limiting how complete a risk picture any model could build
- With only ~11% of respondents flagged, every model struggled with low precision (0.23–0.26) expected when predicting a rare outcome from general survey data, but it meant accuracy alone would have been a misleading way to judge success
- Keeping five separate notebooks (one per model, built independently) consistent same feature set, same leakage checks, same train/test discipline took real coordination across the team



THANK YOU

